

UNIVERSITY OF LIMPOPO

FACULTY OF SCIENCE AND AGRICULTURE
SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES
DEPARTMENT OF COMPUTER SCIENCE

DEGREE AND DIPLOMA MAIN EXAMINATION

OCT/NOV: 2024

MODULE: SCOA032

PAPER: P

(ARTIFICIAL INTELLIGENCE)

TIME: 2 HOURS

MARKS: 60

INTERNAL EXAMINERS: MR VR BALOYI

MR MJD MANAMELA

EXTERNAL EXAMINER: DR O GIWA (NWU)

THIS PAPER CONSISTS OF THREE (03) PAGES INCLUDING COVER PAGE

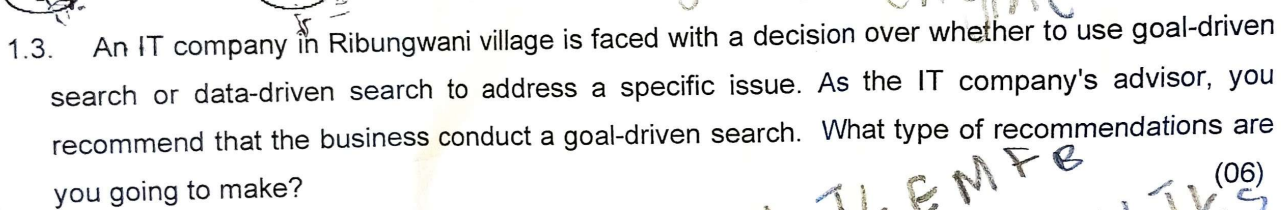
INSTRUCTIONS:

1. Answer all the questions
2. Write neatly and legibly
3. Show all your workings including all the steps.

Finding solutions for Africa

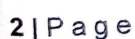
1.1. The Mankweng Hospital requires your knowledge to help its radiologists with computer vision systems and to help them make informed decisions. What kind of computer vision system you can recommend, which elements the system will require, and what current deficits you can include in the recommendation? (10)

(08)



2.1. Use the alpha-beta pruning mechanism to calculate the solution.

(07)



A goal is given in the ^S problem Statement

A large number of rules ~~that~~ ^S matches the condition in the problem statement

Problem data is not given

- 2.2. The production system is a model of computation that has proved particularly important in AI, both for implementing search algorithms and for modeling human problem-solving. A production system provides pattern-directed control of a problem-solving process and consists of a **set of production rules, a working memory, and a recognise act control cycle**. What is the role of the production rules, a working memory, and a recognize-act control cycle? (06)
- 2.3. A production rule is a fundamental unit of knowledge in a production system and consists of two primary components: the condition and the action. What are the roles of Condition-Action in a production rule? (04)

QUESTION 3 [19]

- 3.1. With the help of a diagram, solve the following production rules using forward chaining. (08)

1. $P \wedge Q \rightarrow \text{goal}$ ✓
2. $R \wedge S \rightarrow P$ ✓
3. $X \wedge R \rightarrow Q$
4. $K \wedge U \rightarrow Q$
5. $Z \rightarrow S$ ✓
6. $\text{Start} \rightarrow Z \wedge R \wedge Q$ ✓



Data
default

- 3.2. A well-crafted restaurant script is essential for maintaining consistency in service, ensuring customer satisfaction, and upholding the restaurant's brand identity. By carefully crafting a script for the restaurant, which components should you cover to create a cohesive and professional dining experience for your customers? (05)
- 3.3. Which slots of the frame are represented by the following statements.
- 3.3.1 The "hotel phone" might be a special instance of "phone," which might be an **instance** of a "communication device." (03)
- 3.3.2 An important feature of frames is the ability to attach **procedural** instructions to a slot. (03)

Entry condition
Props
Roles
Scence
Results